

Consider the following problem

$$\min_x \sum_{i \in I} f_i(x_i) \quad s.t. \quad \begin{cases} g_i(x_i) = 0, & i \in \{1, \dots, N\} \\ \sum_{i \in I} A_i x_i = b \end{cases}$$

where $x_i \in \mathbb{R}^{n_x}$, $f_i : \mathbb{R}^{n_x} \rightarrow \mathbb{R}$, $g_i : \mathbb{R}^{n_x} \rightarrow \mathbb{R}^{n_g}$, and $A_i \in \mathbb{R}^{n_A \times n_x}$

Our aim is to exploit the structure of problem to distribute computations on local level as much as possible, as opposed to decomposing the problem,

If we associate multipliers $\lambda_i \in \mathbb{R}^{n_g}$ with g_i and $\mu \in \mathbb{R}^{n_A}$, we will have the following Lagrangian and its derivatives:

$$L(x, \lambda, \mu) = \sum_{i \in I} f_i(x_i) + \lambda_i^T g_i(x_i) + \mu^T (\sum_{i \in I} A_i x_i - b)$$

$$\nabla_x L(x, \lambda, \mu) = \begin{bmatrix} [\nabla_x L(x, \lambda, \mu)]_1 \\ \vdots \\ [\nabla_x L(x, \lambda, \mu)]_N \end{bmatrix}$$

and

$$[\nabla_x L(x, \lambda, \mu)]_i = \nabla_{x_i} f_i(x_i) + \nabla_{x_i} g_i(x_i)^T \lambda_i + A_i^T \mu$$

where ∇g is the Jacobian of the corresponding function.

Therefore to find the minimizer we solve the following problem;

$$\nabla_x L(x, \lambda, \mu) = 0 \quad s.t. \quad \begin{cases} g_i(x_i) = 0, & i \in \{1, \dots, N\} \\ \sum_{i \in I} A_i x_i = b \end{cases}$$

Presuming regularity, and denoting vector of (x_i, λ_i) as z_i , solving this problem using the Newton method will result in the following system of equations.

$$[KKT] \begin{bmatrix} \Delta z \\ \Delta \mu \end{bmatrix} = - \begin{bmatrix} [\nabla_x L(x, \lambda, \mu)]_i \\ g_i(x_i) \\ \vdots \\ \sum_{i \in I} A_i x_i - b \end{bmatrix} \quad (1)$$

Where Δz is a vector of size $N \times (n_x + n_g)$, and $\Delta \mu$ of size n_A with the following KKT matrix;

$$KKT = \begin{bmatrix} \ddots & & & \\ & \begin{bmatrix} \nabla_{x_i}^2 f_i(x_i) + \langle \lambda_i, \nabla_{x_i}^2 g_i(x_i) \rangle & \nabla_{x_i} g_i(x_i)^T \\ \nabla_{x_i} g_i(x_i) & 0 \end{bmatrix} & & \\ & & \ddots & \\ & \dots & A_i & 0 & \dots \end{bmatrix} \begin{bmatrix} \vdots \\ A_i^T \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

The matrix containing Hessian of the Lagrangian andd Jacobians of fucntions g_i 's consists of N blocks of size $(n_x + n_g) \times (n_x + n_g)$ (with the zero matrix being the size $n_g \times n_g$), and the block containing A_i^T has N matrices of size $(n_x + n_g) \times n_A$.

Let us denote the block matrix containing the Hessian of the Lagrangian and the Jacobians as matrix $\mathbf{D}$ and each sub matrix by D_i , the block containing matrices A_i as $\mathbf{A}$, the vector containing i-th component of $\nabla_x L(x, \lambda, \mu)$ and $g_i(x_i)$ as r_1 , and $\sum_{i \in I} A_i x_i - b$ as r_2 in the right hand side of 1. Then we can solve for ΔZ :

$$\begin{aligned} D\Delta z + \mathbf{A}^T \Delta \mu &= -r_1 \\ \Delta z &= -D^{-1}r_1 - D^{-1}\mathbf{A}^T \Delta \mu \end{aligned}$$

Where due to the structure of the KKT matrix with $\Delta \mu$ provided can be solved for each z_i seperately.

$$\Delta z_i = -D_i^{-1}r_{1_i} - D_i^{-1}\mathbf{A}_i^T \Delta \mu$$

And

$$\mathbf{A}\Delta z = -r_2$$

which can be written as

$$\begin{aligned} \sum_{i \in I} \mathbf{A}_i z_i &= -r_2 \\ \sum_{i \in I} -\mathbf{A}_i D_i^{-1}r_{1_i} - \mathbf{A}_i D_i^{-1}\mathbf{A}_i^T \Delta \mu &= -r_2 \\ \sum_{i \in I} -\mathbf{A}_i D_i^{-1}\mathbf{A}_i^T \Delta \mu &= -r_2 - \sum_{i \in I} \mathbf{A}_i D_i^{-1}r_{1_i} \end{aligned}$$

where the terms with index i will be calculated locally and communicated for finding Lagrange multiplier of the coupled constraint. The updated $\Delta \mu$ will be communicated to local agents to solve for Δz .

For example consider the following:

$$\min_x \sum_{i \in I} \frac{1}{2} x_i^T Q_i x_i \quad s.t. \quad \begin{cases} A_i x_i = b_i, & i \in \{1, \dots, N\} \\ \sum_{i \in I} C_i x_i = c \end{cases}$$

where $x_i \in \mathbb{R}^{n_x}$, $Q_i \in \mathbb{R}^{n_x \times n_x}$, $A_i \in \mathbb{R}^{n_A \times n_x}$, and $C_i \in \mathbb{R}^{n_C \times n_x}$. We will have the following Lagrangian and its gradient.

$$L(x, \lambda, \mu) = \frac{1}{2} \sum_{i \in I} x_i^T Q_i x_i + \lambda_i^T (A_i x_i - b_i) + \mu^T \left(\sum_{i \in I} C_i x_i - c \right)$$

$$\nabla_{x_i} L(x, \lambda, \mu) = Q_i x_i + A_i^T \lambda + C_i^T \mu$$

By denoting (x_i, λ_i) as z_i we have the following system of equations:

$$\left[\begin{array}{ccc} \ddots & & \\ & \begin{bmatrix} Q_i & A_i^T \\ A_i & 0 \end{bmatrix} & \\ & & \ddots \\ \dots & C_i & 0 & \dots \end{array} \right] \begin{bmatrix} \vdots \\ C_i^T \\ 0 \\ \vdots \\ 0 \end{bmatrix} \begin{bmatrix} z \\ \mu \end{bmatrix} = \begin{bmatrix} \vdots \\ 0 \\ b_i \\ \vdots \\ c \end{bmatrix}$$

If we denote the each blocks matrices $[Q_i, A_i^T; A_i, 0]$ (with zero block being size $n_A \times n_A$) as F_i and the whole matrix as $\mathbf{F}$, the vector $[0; b_i]$ (zero vector is of size n_x) as B_i and the vector of that collects them as B , denote $[C_i, 0]$ (zero matrix of size $n_A \times n_C$) as $\bar{C}_i$, and the matrix containing them as $\bar{\mathbf{C}}$. Then we will have;

$$\mathbf{F}z + \bar{\mathbf{C}}^T \mu = B$$

$$z = \mathbf{F}^{-1}B - \mathbf{F}^{-1}\bar{\mathbf{C}}^T \mu$$

$$z_i = \mathbf{F}_i^{-1}B_i - \mathbf{F}_i^{-1}\bar{\mathbf{C}}_i^T \mu \quad (2)$$

$$\bar{\mathbf{C}}z = c$$

$$\sum_{i \in I} \bar{\mathbf{C}}_i z_i = \sum_{i \in I} \bar{\mathbf{C}}_i (\mathbf{F}_i^{-1}B_i - \mathbf{F}_i^{-1}\bar{\mathbf{C}}_i^T \mu) = c$$

$$\sum_{i \in I} \bar{\mathbf{C}}_i \mathbf{F}_i^{-1} \bar{\mathbf{C}}_i^T \mu = \sum_{i \in I} \bar{\mathbf{C}}_i \mathbf{F}_i^{-1} B_i - c \quad (3)$$

Here again the terms with index i are calculated locally and then communicated to central node to find μ , which will be sent to local nodes to solve for z_i .

Which is solved by the following algorithm.

Algorithm 1 Decomposition

Input: Problem parameters Q_i, A_i

Output: decomposition of F_i, dF_i

- 1: $\forall i$: Assemble matrices F_i
 - 2: Find LU decomposition of matrices F_i as dF_i
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Algorithm 2 Assemble equation for coupled Lagrange multiplier 1

Input: dF_i , $\tilde{C}_i$, and B_i

Output: $\mathbf{F}_i^{-1}\tilde{\mathbf{C}}_i^T$, $\tilde{\mathbf{C}}_i\mathbf{F}_i^{-1}\tilde{\mathbf{C}}_i^T$, $\mathbf{F}_i^{-1}B_i$, and $\tilde{\mathbf{C}}_i\mathbf{F}_i^{-1}B_i$

- Assemble matrix $\mathbf{F}_i^{-1}\tilde{\mathbf{C}}_i^T$
 - 2: Assemble matrix $\tilde{\mathbf{C}}_i\mathbf{F}_i^{-1}\tilde{\mathbf{C}}_i^T$
 - Assemble matrix $\mathbf{F}_i^{-1}B_i$
 - 4: Assemble matrix $\tilde{\mathbf{C}}_i\mathbf{F}_i^{-1}B_i$
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Algorithm 3 Basic distributed algorithm for linear quadratic problem

Input: Problem parameters Q_i , A_i , b_i , C_i , and c

Output: z_i , and μ

- $\forall i$: Compute decomposition dF_i using algorithm 1
 - $\forall i$: Assemble matrices using algorithm 2
 - 3: Solve for μ in 3
 - Solve for z_i in 2
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